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Enhancing Dybbølsbro Bridge – Refurbishment and expansion without new substructure

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ABSTRACT: The refurbishment and expansion of the Dybbølsbro Bridge in Copenhagen presents a unique engineering challenge. This contribution details the process of suspending a new 2000 m^2 bridge between two existing bridges, all while maintaining traffic on the existing bridges, and executing work over twelve operational railway tracks. The innovative expansion uses a deck superstructure that requires no additional substructure. Achieved by strengthening and utilizing existing bridges, the new bridge consists of 55 independent steel elements. By reusing the existing bridges for the new superstructure and avoiding construction of a new substructure, we minimized the material consumption, consequently reducing the project's environmental footprint and reduced the disturbance of the rail traffic, all contributing to a more sustainable project. The expansion of the bridge epitomizes the future of sustainable urban infrastructure, demonstrating that conscientious planning and innovative design can bring ambitious infrastructure projects to fruition, enhancing city landscapes and serving communities better.

1 INTRODUCTION

Dybbølsbro, a pivotal structure in Copenhagen's urban landscape, has undergone a significant transformation. Originally erected around 1900 as a steel truss bridge, initially served as a vital connection over the Copenhagen Freight Yard to Kalvebod Brygge. The bridge was replaced by pedestrian and cycle path bridge in the mid 1980's. In mid-1990's a parallel road bridge was erected creating a ~10 m wide gap between the two 190 m long bridges. Both existing bridges were push launched and constructed as post tensioned 5 span concrete bridges.

This paper presents an in-depth exploration of the recent expansion of Dybbølsbro, a project that seamlessly integrates contemporary engineering excellence with the bridge's storied past. The expansion initiative is a testament to Copenhagen's commitment to sustainable urban development and the enhancement of public infrastructure. The undertaking addressed the need for increased capacity and improved safety for pedestrians and cyclists.

The Dybbølsbro expansion project, characterized by its innovative approach represents a paradigm shift in urban bridge expansion. Challenges, such as the integration of new structures with the existing framework and minimizing disruption to the city life, were met with advanced engineering solutions. The project's approach to utilizing the existing concrete structures, coupled with the implementation of state-of-the-art digital tools, sets a new benchmark in bridge engineering and urban infrastructure development.

As urban spaces evolve, the demand for infrastructure that can adapt and accommodate growing needs becomes more important. The Dybbølsbro project emerges as a beacon of such adaptability, showcasing how existing structures can be repurposed to serve new requirements.

By examining the Dybbølsbro expansion project, this paper aims to provide valuable insights into the complexities and opportunities of modernizing existing infrastructure.

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2 PROJECT OVERVIEW AND TECHNICAL APPROACH

The bridges are situated in a high-traffic zone, accommodating a steady stream of pedestrians, cyclists, and vehicles. They span across all railway tracks that feed into Copenhagen's central station. It was crucial that throughout the construction process, these varied modes of transport could continue to operate, ensuring minimal disruption during construction.

2.1 *Existing conditions*

Dybbølsbro, comprised of continuous spans in post-tensioned concrete box girders, approx. 40 m in span and totaling approx. 190 meters in length, rests on pile foundations embedded in old seabed. The existing bridges, launched from the eastern end, had surplus load-bearing capacities due to their robust prestressing and erection scheme. This structural reserve played a crucial role in the expansion project, allowing for the resistance of additional loading.

The two existing bridges have a gap of approx. 10.2 m between the edge beams, 14.4 m between the faces of the box girders and approx. 19 m between c/c of substructure which is also sketched in Figure 1.

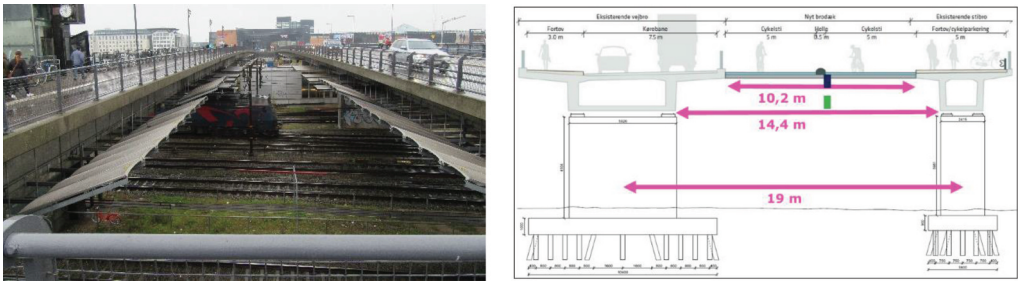


Figure 1. Photo of existing bridges and underpassing railway tracks (left) and main dimensions between bridges (right).

2.2 *External project interfaces and neighboring projects*

The bridges are located at an area giving a lot of different external interfaces which include:

- Underpassing railway track leading into Copenhagen central station
- Catenary system for railway tracks
- A train station having access from the pedestrian bridge
- High number of cyclists and pedestrians crossing the bridge
- A nearby area for long distance busses which had most busses cross the road bridge
- Road traffic on the bridge
- Neighboring projects including plans for erecting twin high-rise student accommodation houses and a large warehouse.

All these interfaces had to be considered and handled during design and under construction. Several of these interfaces heavily influenced the final design of the bridge, in particular the underpassing railway.

2.3 *Considered concepts*

Suspending a new steel bridge between the existing concrete structures required a comprehensive understanding of the existing bridges' load-bearing capabilities and the intricacies involved in integrating a new structure.

Critical to the project was the analysis of static systems and the effective distribution of loads in relation to the existing load carrying capacities. The existing bridges, being post-tensioned continuous spans on pile foundations, offered a robust platform for expansion. Leveraging the existing concrete structures was a cornerstone of this project. The existing bridges were originally designed with surplus load-bearing capacities since both bridges were

push-launched during construction. The design philosophy focused on utilizing these reserves to support the additional weight of the new bridge.

The strategy for integrating a new steel bridge involved suspending it between the existing structures. This approach was chosen for its minimal impact on the existing bridges. The suspension system was to be engineered to ensure stability and safety, with careful attention to the load distribution to avoid overloading any part of the existing structures.

The bridge widening concept needed to consider all the external boundaries and limitations while still resulting in an effective expansion of the available bridge deck area. Initially four different alternatives were considered. These four include:

- New steel deck suspended between wings on existing bridge decks (top left in Figure 2)
- New steel deck suspended between boxes, i.e. 14.4 m spans (top right in Figure 2)
- Beams between existing columns, deck spanning longitudinally between beams, i.e. 14.4 m or 19 m spans to avoid eccentric load on substructure (bottom left in Figure 2)
- New independent parallel bridge (bottom right in Figure 2)

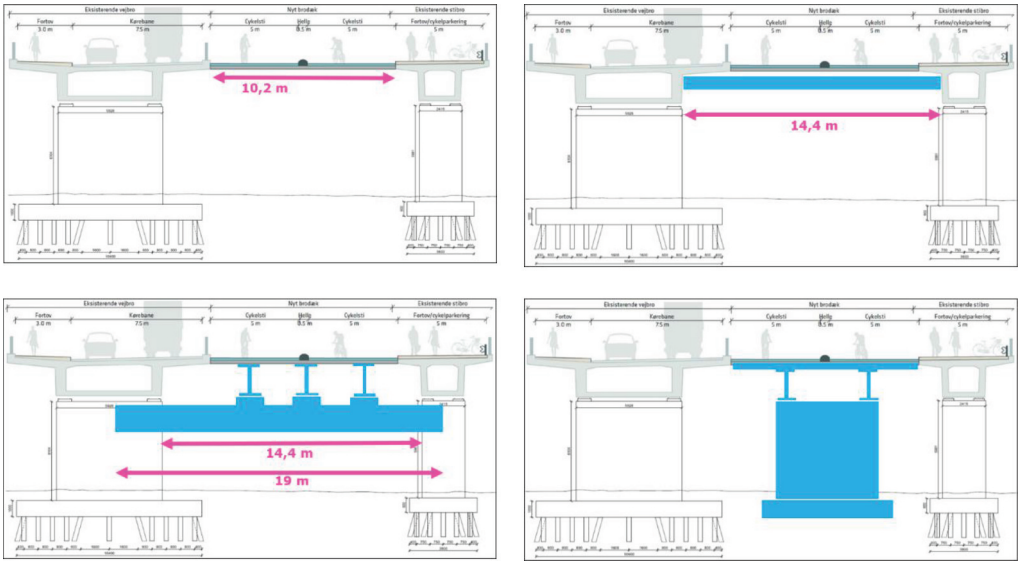


Figure 2. The four concepts initially considered.

All four alternatives had their pros and cons. Alternative 1 with deck suspended between wings on existing bridges was chosen due to the advantages of low self-weight, span width of new structure minimized and simple installation of bridge elements. This alternative further had the disadvantage of high loading on bridge wings and high eccentric loading of existing box girders and substructures. These negative effects were reduced through clever detailing of the permanent supports for the steel deck and implementation of structural parts to reduce the eccentric loading on the substructure. This is further described in the following sections.

2.4 *Design philosophy and structural system integration of implemented concept*

The core challenge of the implemented alternative was to increase the bridge area by 70% without constructing new supports. The design philosophy revolved around leveraging the existing structures' strengths, particularly their load-bearing reserves. The chosen approach was to suspend a new steel bridge between the existing concrete bridges, a method that presented both structural and construction advantages. This approach required detailed planning and precision to ensure compatibility with the existing structures.

Analyzing the static systems and load distribution capabilities of the existing structures was essential. The expansion involved the careful calculation of loads and the distribution of

additional weight between the existing structures. The static system for the new steel bridge was designed to minimize eccentric loading on the existing bridges. The engineering team explored multiple alternatives, eventually selecting a system that optimally distributed the loads while ensuring structural integrity and safety.

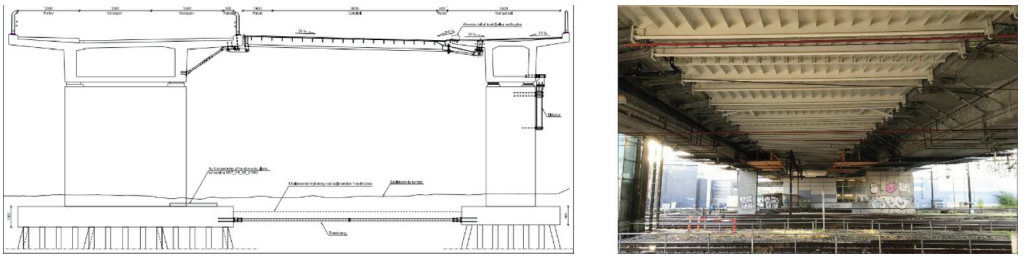


Figure 3. Cross section (left) and photo (right) of bridge after installation of suspended steel deck.

Identification of weaknesses of the existing bridges were essential. Some weaknesses required strengthening of the bridges while others were not possible to strengthen, thus required alternative ways of resisting external forces.

The most significant change due to a weakness was the change in the overall statical system of the three combined bridges. Suspending a deck between the existing bridges results in torsion in the superstructure leading to transverse bending in the substructure. This was, in particular, critical for the pedestrian bridge. The transverse bending was instead resisted by horizontally paired forces where the new steel deck was used as a compression member while new tension members connected the pile caps of the existing concrete bridges.

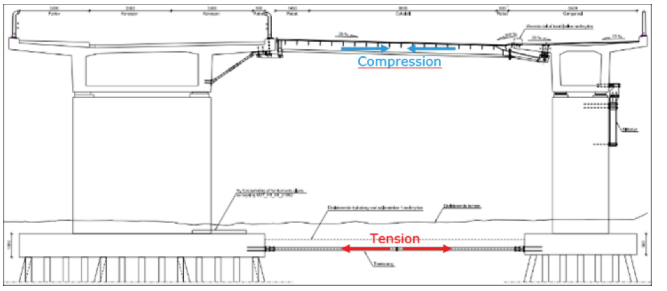


Figure 4. Compression and tension members between existing bridges resists the transverse bending introduced by the eccentric loading created by the suspended steel deck.

3 TECHNICAL CHALLENGES AND SOLUTIONS

The project encountered several technical challenges, notably the need to work above active railway tracks and maintain uninterrupted traffic flow. The solutions included construction of the bridge in elements that could be installed during track possessions of typically 2-3 tracks during nighttime, which allowed for efficient and safe work without significant disruption of the traffic above and below the bridge. This section elaborates on some of the challenges handled in order to meet the requirements to maintain road, pedestrian and cycle traffic and train operation during construction.

3.1 *Safety measures over operational railway tracks*

Twelve railway tracks pass under the bridge. All tracks have very high traffic intensities. Major closures over longer periods were not an option. Track possessions of typically 2-3 tracks during a night were allowed. This called for phased construction by constructing the bridge of

individual elements. Carrying out the full construction of the bridge necessitated approx. 60 instances of track possessions distributed over the approx. 9 months construction period.

Working above active railway tracks presented safety challenges. The steel elements were attached to the bottom of the existing superstructure wings. It was therefore of high importance that unhindered access to the bottom face of the bridges was given. The access was provided by temporary platforms attached to the webs of the box girders. These were installed in segments during nightly track possessions. The platforms included shielding towards the underpassing railway tracks making it possible to work without any disturbance from the railway traffic.



Figure 5. Access platforms below the bridge superstructures.

3.2 *Effects on existing bridges*

Suspending the steel deck directly between the two inner edge beams would introduce torsion into the bridge decks and result in transverse bending in the substructures. This was not a problem for the road bridge but turned out to be critical for the pedestrian bridge. The issue on the pedestrian bridge was solved by connecting the deck closer to the webs of the superstructure as shown in Figure 6.

To ensure sufficient bearing capacity of the wing on the road bridge superstructure, compression struts were introduced as shown in Figure 6. These ensured that the allowable vehicle loading on the bridge could remain the same after the widening. Combined, the 2 solutions are a clear example on how different solutions can be required to meet the different requirements from the existing structures.

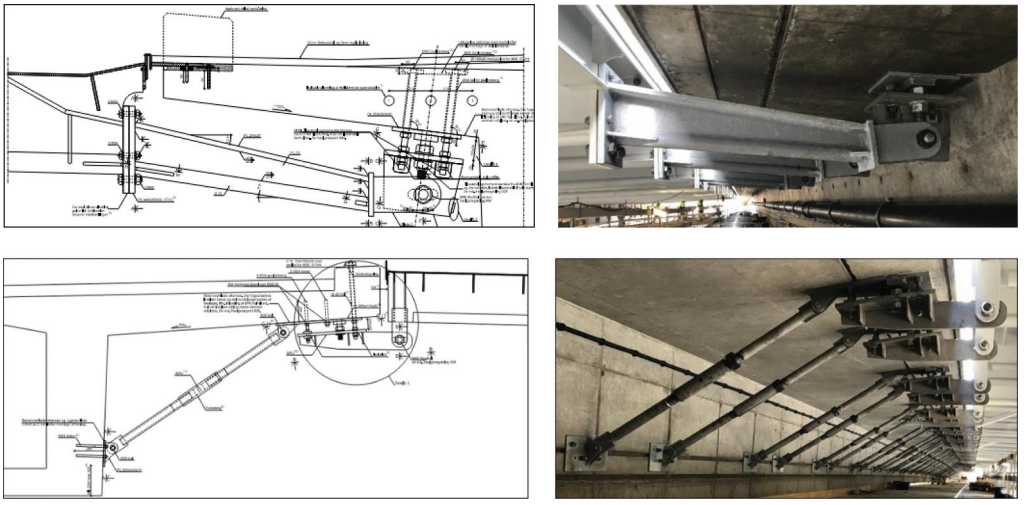


Figure 6. Upper left and right: Suspension of steel deck on pedestrian bridge. Lower left and right: Compression struts below wing on road bridge.

3.3 *Element installation and assembly*

Typically, three elements were installed during a night. Effective installation was required especially for elements above railway tracks since reopening of the railway traffic after a nightly track possession was mandatory. Lifting brackets were developed serving a dual purpose meaning they could be used for lifting the elements into position while also acting as temporary supports for the elements until the permanent supports were installed. After the elements had been placed on the temporary supports it was possible to move the elements into their final position before installation of permanent supports.

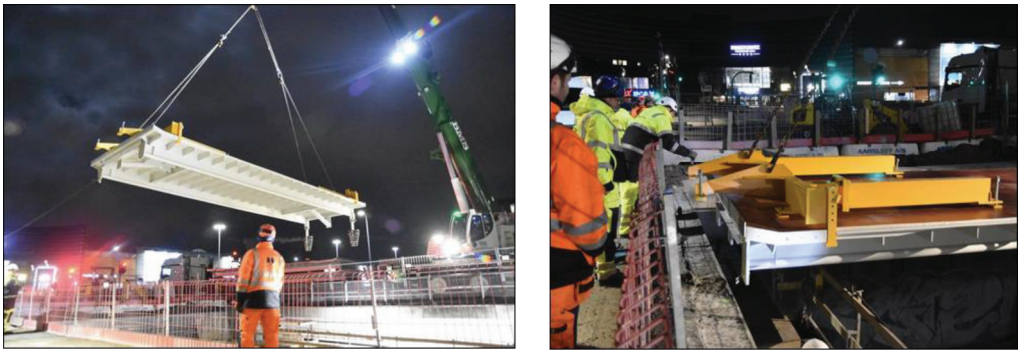


Figure 7. Lifting brackets also serve the purpose as temporary element support.

The steel elements required joining to guarantee the durability of the surfacing and the cyclists' comfort. The temporary platforms below the existing bridges gave access to the existing bridges' wings and approximately 20% of the area of the steel elements. However, as there was no access to the remaining roughly 80% of the steel elements' underside, a bolted connection assembled from above was designed. This connection's primary function was to assure that neighboring elements would deflect simultaneously. The plate in the bolt connection had oversized holes to take up any tolerances during installation. The bolt connection is shown in Figure 8.



Figure 8. Connecting the steel elements from above on the central part of the spans.

3.4 *Permanent supports for steel elements*

The two existing concrete bridges were in theory parallel and shared the same longitudinal profile. However, there were small deviations due to construction tolerances etc. which meant the need for tolerances for installation of the new steel elements. The majority of the tolerances were handled in the permanent support brackets. These were designed to take up tolerances of ±30 mm in transverse and longitudinal directions of the bridge and ±20 mm vertically. The ability to take up tolerances meant that installation of the new steel deck could be carried out without any requirements for demolition and adaption of old/new structure.



Figure 9. Permanent support brackets with capacity for taking up tolerances. Bracket on pedestrian bridge (left) and road bridge (right).

In conclusion, the technical approach to the Dybbølsbro expansion project exemplifies engineering ingenuity. The careful integration of a new steel structure with the existing bridges and the solution-oriented approach to challenges contributed to the project's success.

4 SUSTAINABILITY AND ENVIRONMENTAL CONSIDERATIONS

The expansion of Dybbølsbro not only focused on engineering innovation but also heavily emphasized sustainability. This approach reflects Copenhagen's broader commitment to sustainable urban development.

One of the project's primary sustainability strategies was the reuse of the existing bridge structures. This approach significantly reduced the material consumption typically associated with bridge expansion projects. By leveraging the existing concrete bridges' load-bearing reserves, the need for new substructures was eliminated, thereby minimizing the use of new raw materials and reducing the project's carbon footprint.

Widening the bridge deck without construction of a new substructure is also sustainable in the sense that it significantly reduced the disturbance of the underpassing railway tracks and reduced the economic range for the project.

The project's sustainability considerations extended beyond the construction phase. The new bridge was designed with sustainability in mind, ensuring that future maintenance would be efficient and environmentally friendly. Future surface treatment of the steel deck bottom face towards the railway tracks will have a major impact on the underpassing railway. Therefore, the steel elements were given the best possible surface treatment which included a duplex painting system C5VH. This included spray metallization and paint.

In conclusion, the Dybbølsbro expansion project stands as an example of sustainable engineering, balancing the demands of modern infrastructure development with environmental stewardship.

5 DIGITAL TOOLS AND TECHNOLOGY

The Dybbølsbro expansion project represents a forward-thinking approach to bridge engineering, leveraging advanced digital tools and technologies, all playing an important role in ensuring the project's precision, efficiency and success.

Scanning for Precise Expansion One of the technological highlights was the use of 3D Light Detection and Ranging (LiDAR) scanning. This tool provided three-dimensional data of the existing bridge structures, crucial for the design and integration of the new steel bridge. The LiDAR scans offered detailed insights into the geometry of the existing structures, enabling engineers to make informed decisions about the design and placement of the new components. The information made sure that tolerances required in the project in order for the new deck to be installed on the existing bridges could be determined. This ensured that installation of the elements and permanent supports could be carried out without any major challenges.

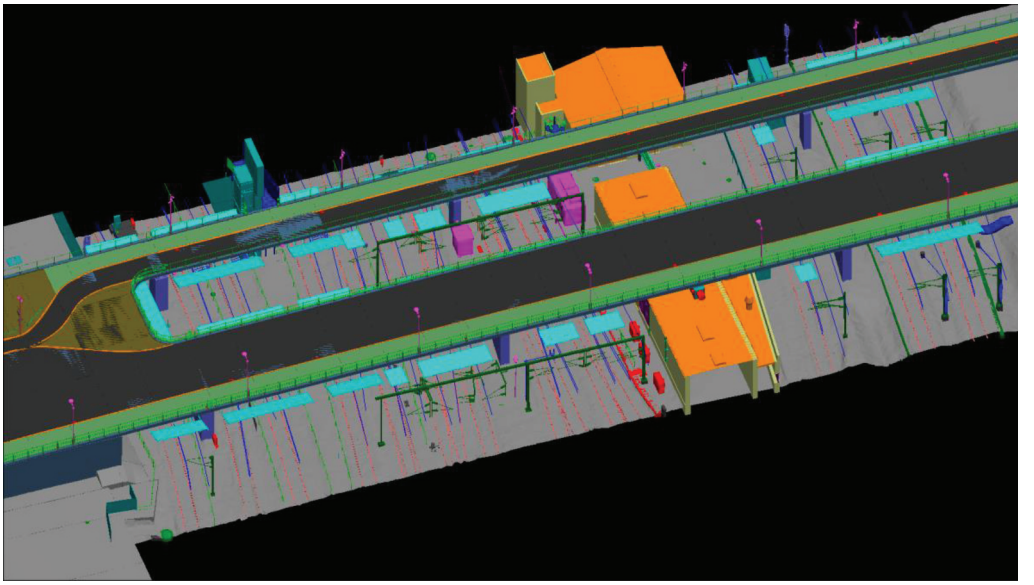


Figure 10. 3D-model created from 3D-LiDAR scanning of project area before installation of the new steel deck. Approximately half of the bridge length is shown.

6 RESULTS AND CONCLUSION

The Dybbølsbro expansion serves as a model for urban infrastructure projects worldwide. It demonstrates how engineering ingenuity, combined with a focus on sustainability and community needs, can lead to significant urban improvements. The project's ability to balance existing structure preservation with modern infrastructure needs is particularly noteworthy.

The project successfully expanded the bridge without constructing a new substructure, an innovative feat that required meticulous planning and execution. A notable accomplishment of the expansion is the creation of Denmark's widest bicycle track. The expanded bicycle track offers ample space for cyclists and pedestrians, significantly improving safety and encouraging more people to opt for cycling over motorized transport.

The expansion project placed a strong emphasis on sustainability, evident in the reuse of existing structures. This approach not only reduced the environmental impact but also preserved the existing structures at Dybbølsbro.